

Adaptive Bandit-Guided Tabu Search for Last-Mile Delivery Routing in Dammam

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Abstract—Efficient last-mile delivery routing requires route plans that remain feasible under capacity, service-time, time-window, and congestion constraints. This study develops Adaptive Bandit Tabu Search with Roulette Warm Start (ABTS-RW) for a realistic capacitated vehicle routing problem with time windows (CVRPTW) in Dammam, Saudi Arabia. A benchmark dataset is constructed from OpenStreetMap road-network and delivery point-of-interest (POI) data, with customer demand, service times, time windows, and five congestion scenarios. ABTS-RW first applies roulette-guided neighborhood search for early diversification and then uses a multi-armed-bandit controller to select Tabu Search (TS) neighborhoods during intensification. In the 100 run robustness comparison on 150 and 200 customer instances, ABTS-RW improves over TS by 3.81% on average, wins 94 out of 100 paired comparisons, and increases feasible-run coverage from 87/100 to 98/100. Non-parametric tests and bootstrap confidence intervals confirm that the improvement is statistically reliable.

Index Terms—Capacitated vehicle routing problem, last-mile delivery, Tabu Search, multi-armed bandit, adaptive metaheuristics.

I. INTRODUCTION

In recent years, customer expectations have shifted toward faster, more flexible, and more reliable delivery services. As a result, last-mile delivery has become one of the most important and costly stages of urban logistics, directly affecting delivery time, service reliability, operating cost, fleet utilization, and environmental impact [1]. It can account for more than 40% of total supply chain costs [2]. Last-mile logistics has also been estimated to account for nearly 43% of the CO₂-equivalent emissions generated across the fulfillment process [3]. Urban freight interventions have been associated with average reductions of 25% in cruising, 41% in last-mile emissions, and 28% in delivery times [4].

This study focuses on last-mile delivery routing in Dammam, Saudi Arabia. Primary objective of this work is to design routes that start from a depot that will serve geographically distributed customers while satisfying vehicle-capacity and customer time-window restrictions while reducing total driver-duty time. Contrary to simplified routing models, practical delivery routing depends on road-network structure, congestion, service times, waiting time, and delivery-window pressure [5].

The problem is modeled as a Capacitated Vehicle Routing Problem with Time Windows (CVRPTW) that extends the Capacitated Vehicle Routing Problem (CVRP) by requiring service at each customer to begin within an allowable time interval [5]. While this extension improves operational realism, increases the optimization difficulty assignment, sequencing, travel time, service time, waiting time, and lateness must be considered together [6]. Vehicle routing is a computationally hard combinatorial optimization problem, and the number of possible routes grows rapidly with the number of customers [7]. Capacity constraints tightly connect customer assignment with route construction, and road-network travel times make realistic urban routing even more complex than Euclidean-distance routing [1], [3]. Time windows and service times add scheduling pressure, and congestion scenarios can change the difficulty of the same customer set [5].

This study develops and evaluates an adaptive heuristic method for a realistic CVRPTW benchmark dataset. Only a few studies combine realistic road-network & aforementioned concerns with the time-window pressure in one study as most of the routing studies rely solely on standard benchmark instances [8]. Even though strong classical heuristics are competitive till this date, performances depends heavily on neighborhood selection, diversification, and intensification [9]. The proposed Adaptive Bandit Tabu Search with Roulette Warm Start (ABTS-RW) combines a short roulette-guided diversification phase with adaptive multi-armed-bandit (MAB) Tabu Search (TS) intensification to address this.

The remainder of this study is organized as follows: section II reviews related work, section III presents the problem formulation and modeling approach, section IV describes the proposed ABTS-RW method and comparison algorithms, section V reports the experimental evaluation, and Section VI concludes the study with recommendations for future work.

II. LITERATURE REVIEW

The literature on vehicle routing covers exact optimization, constructive heuristics, classical metaheuristics, and learning-assisted search. Exact and constraint-programming methods provide rigorous modeling frameworks, but their computational burden soars sharply when practical constraints such

as capacity limits, time windows, service times, and realistic travel-time matrices are included [1].

Constructive and local-improvement heuristics such as greedy insertion, nearest-neighbor construction, and swap neighborhoods form the simplest class of vehicle routing methods. Tiwari and Sharma [1] compared greedy construction, local search (LS), and TS for last-mile routing and showed that simple heuristics can be useful for smaller or less constrained cases. However, their solution quality usually drops as customer size, constraints, and operational realism increase.

Classical metaheuristics are designed to provide stronger exploration. For instance, TS uses memory to avoid cycling and guide local search toward new solution regions [10], Simulated Annealing (SA) accepts selected worsening moves to escape local optima, while Variable Neighborhood Search (VNS) and Iterated Local Search (ILS) improve solutions through perturbation and neighborhood changes [9]. Hybrid Genetic Search (HGS) has also shown strong CVRP benchmark performance and has benefited from open-source implementation work [11]. These methods are effective, but their performance mostly depends on neighborhood design, parameter settings, and the balance between diversification and intensification.

Adaptive Large Neighborhood Search (ALNS) is widely used because it can handle rich constraints by modifying repair logic and acceptance rules. Özarić et al. [12], applied ALNS to last-mile deliveries with stochastic customer availability and multiple visits. Nevertheless, ALNS performance can depend strongly on the selected destroy-repair portfolio, operator tuning, and acceptance policy.

Learning-based and hybrid methods have become increasingly relevant. Kalatzantonakis et al. [13] combined reinforcement learning (RL) with VNS for the CVRP and showed that learning can guide neighborhood selection while optimization maintains feasibility. Fitzpatrick et al. [7] proposed a scalable learning-based approach for large routing instances, Ramírez-Villamil et al. [14] combined machine learning with routing optimization for parcel delivery. Mesa et al. [15] used a data-driven Greedy Randomized Adaptive Search Procedure (GRASP) to predict last-mile delivery sequences. These studies suggest that learning is most useful when it supports an optimization procedure rather than replacing it. However, learning-based methods can face challenges in training data, generalization, reproducibility, and fair comparison with strong classical baselines.

Multi-armed bandit (MAB) offers a lightweight and interpretable approach for adaptive search. Contextual MABs have also been applied to sequential decision problems such as personalized itinerary planning, where decisions depend on route and user context [16]. In routing, compared with end-to-end learning, bandit-guided search is more attractive because it does not require large training datasets and can update decisions online during optimization.

Table I compares the main routing method families using criteria that are relevant to this study.

TABLE I
CRITERIA-BASED COMPARISON OF ROUTING METHOD FAMILIES

Criterion	Constructive/local heuristics	Metaheuristics	Adaptive/hybrid methods	Representative examples
Main role	Fast route construction and local repair	Stronger exploration and intensification	Adaptive operator selection and learning-guided search	Greedy insertion, 2-opt, TS, SA, VNS, ALNS, RL-guided search
Exploration ability	Low to medium	High	High	SA, VNS, ILS, ALNS
Adaptive control	Low	Medium	High	ALNS, RL-guided search, bandit-guided search
Feasibility handling	Moderate; depends on repair logic	High when penalties and repair are designed carefully	High when feasibility-aware evaluation is retained	TS, ALNS, ILS/VNS
Scalability and realism	Often limited on rich instances	Good but parameter-sensitive	Good when tuned and evaluated under common protocols	HGS, learning-assisted CVRP, public-road studies
Fit to this study	Baseline and initialization role	Strong classical comparison role	Main proposed direction	Constructive, Tabu Search, SA, ILS/VNS, ALNS, Proposed ABTS-RW

III. PROBLEM FORMULATION AND MODELING

A. Benchmark Dataset Construction

The routing task is modeled as a single-depot CVRPTW on a directed road-network graph [13]. Vehicles depart from a depot, serve geographically distributed customers, satisfy capacity and time-window restrictions, and return to the depot [6].

NOMENCLATURE

$G = (V, E)$	Directed road-network graph
V	Set of nodes, including depot and customers
E	Set of directed arcs
0	Depot node
C	Set of customer nodes
K	Set of available vehicles
Q	Vehicle capacity
d_i	Demand of customer i
s_i	Service time of customer i
$[a_i, b_i]$	Time window of customer i
t_{ij}^ω	Travel time from node i to node j under scenario ω
x_{ijk}	Binary variable equal to 1 if vehicle k travels from i to j
y_{ik}	Binary variable equal to 1 if customer i is served by vehicle k
A_i	Arrival time at customer i
B_i	Service start time at customer i
W_i	Waiting time at customer i
L_i	Lateness at customer i
D_k	Driver-duty duration of route k
V_k	Capacity violation of route k
S	Complete routing solution
$M(S)$	Number of missing customers in solution S
$U(S)$	Number of duplicate customer visits in solution S
$\lambda_L, \lambda_Q, \lambda_M, \lambda_U$	Penalty weights for infeasibility components

B. Objective Function

For a route assigned to vehicle k , the driver-duty duration is computed as the sum of travel time, service time, and waiting time:

$$D_k = \sum_{i \in V} \sum_{\substack{j \in V \\ j \neq i}} t_{ij}^\omega x_{ijk} + \sum_{i \in C} s_i y_{ik} + \sum_{i \in C} W_i y_{ik} \quad (1)$$

The complete objective function is defined as:

$$\begin{aligned} \min F(S) = & \sum_{k \in K} D_k + \lambda_L \sum_{i \in C} L_i \\ & + \lambda_Q \sum_{k \in K} V_k + \lambda_M M(S) + \lambda_U U(S) \end{aligned} \quad (2)$$

The first term minimizes total driver-duty time. The remaining terms penalize lateness, capacity violation, missing customers, and duplicate customer visits.

C. Constraints

The routing plan must satisfy customer-service, depot-flow, capacity, and time-window constraints.

- Customer service constraints: Each customer must be served exactly once:

$$\sum_{k \in K} y_{ik} = 1, \forall i \in C \quad (3)$$

The incoming and outgoing route arcs must be consistent with the vehicle assignment:

$$\sum_{\substack{j \in V \\ j \neq i}} x_{ijk} = \sum_{\substack{j \in V \\ j \neq i}} x_{jik} = y_{ik}, \forall i \in C, \forall k \in K \quad (4)$$

- Depot-flow constraints: Each active vehicle must leave the depot and return to the depot at most once:

$$\sum_{j \in C} x_{0jk} = \sum_{i \in C} x_{i0k} \leq 1, \forall k \in K \quad (5)$$

- Capacity constraints: The total demand assigned to each vehicle must not exceed the vehicle capacity:

$$\sum_{i \in C} d_i y_{ik} \leq Q, \forall k \in K \quad (6)$$

In the computational evaluator, any excess load is recorded as a capacity violation V_k and penalized in the objective function.

- Time propagation constraints: If vehicle k travels from customer i to customer j , the arrival time at j must account for service time at i and scenario-specific travel time:

$$\begin{aligned} A_j \geq & B_i + s_i + t_{ij}^\omega - M_0(1 - x_{ijk}), \\ & \forall i, j \in V, i \neq j, \forall k \in K \end{aligned} \quad (7)$$

where M_0 is a sufficiently large constant.

- Waiting-time and time-window constraints: Service can start only after arrival and after the opening of the customer time window:

$$B_i \geq A_i, \forall i \in C \quad (8)$$

$$B_i \geq a_i, \forall i \in C \quad (9)$$

Waiting time is computed as

$$W_i = B_i - A_i, \forall i \in C \quad (10)$$

Lateness is defined as the amount by which service starts after the closing of the time window:

$$L_i \geq B_i - b_i, \forall i \in C \quad (11)$$

$$L_i \geq 0, \forall i \in C \quad (12)$$

- Binary restrictions:

$$x_{ijk} \in \{0, 1\}, \forall i, j \in V, i \neq j, \forall k \in K \quad (13)$$

$$y_{ik} \in \{0, 1\}, \forall i \in C, \forall k \in K \quad (14)$$

D. Modeling Approach and Assumptions

The model uses deterministic scenario-specific travel-time matrices rather than real-time traffic updates. Congestion is represented through five fixed operating scenarios. The base scenario is the reference condition with a mean travel-time factor of 1.00. The midday, morning peak, evening peak, and pessimistic scenarios increase the mean travel-time factor to 1.14, 1.38, 1.42, and 1.67, respectively. The fleet is homogeneous, all vehicles have the same capacity Q , and all routes start and end at the same depot. Customer demands are integer quantities, while service times and time windows are assigned at the customer level. Split deliveries, stochastic cancellations, heterogeneous vehicles, and dynamic order arrivals are not modeled.

IV. PROPOSED SOLUTION

The proposed method, ABTS-RW, is a two-phase online-learning-guided adaptive metaheuristic that combines short diversification with adaptive intensification. In the final configuration, 10% of the runtime is assigned to roulette-guided warm-start search, and the remaining 90% is assigned to MAB controlled TS.

The computational pipeline has four stages. First, a capacity-aware and time-window-aware constructive procedure builds an initial solution. Second, roulette-guided search diversifies this solution by sampling route-improvement neighborhoods. Third, MAB-controlled TS adaptively selects neighborhoods and improves the solution using Tabu memory. Finally, the best solution is selected using the CVRPTW evaluator and a feasibility-first ranking rule.

A. ABTS-RW Method

The overall ABTS-RW framework is shown in Fig. 1.

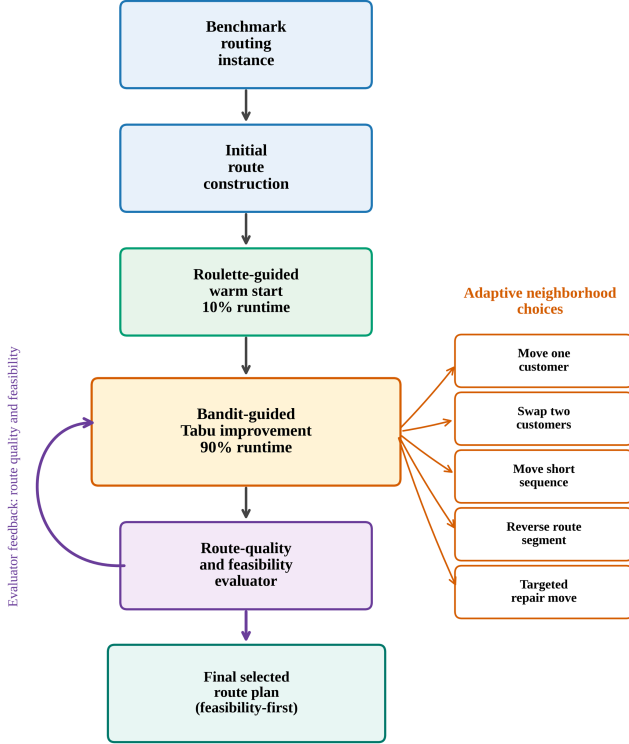


Fig. 1: Proposed ABTS-RW framework.

The roulette warm-start phase provides early exploration before the main Tabu intensification phase. During this phase, neighborhood operators are selected probabilistically. Operators with better recent performance receive higher probability, but weaker operators can still be sampled.

After the warm-start phase, the method switches to MAB-controlled TS. Each routing neighborhood is treated as a bandit arm. At each iteration, the bandit policy selects an operator using its reward estimate and an exploration term. Candidate moves are then generated and evaluated. Recently reversed moves are restricted by Tabu memory unless an aspiration condition is satisfied. Operator rewards are updated using search feedback such as objective improvement, feasibility recovery, and preservation of high-quality solutions. The best feasible solution, or the best penalized solution when no feasible solution is available, is retained throughout the search.

The MAB-Tabu operator set includes relocate-1, swap-1, Or-opt-2, intra-route 2-opt, and targeted relocate neighborhoods. These operators support customer movement between routes, customer exchange, short sequence relocation, intra-route reordering, and targeted repair of weak or infeasible route segments.

Proposed ABTS-RW Procedure algorithm is provided below:

Algorithm 1. Proposed ABTS-RW Procedure

1. Build a capacity-aware and time-window-aware initial route plan.
2. Apply roulette-guided warm-start search for 10% of the runtime.
3. Initialize Tabu memory and bandit statistics for the neighborhood operators.
4. While runtime remains, select a neighborhood using the bandit policy.
5. Generate candidate moves and reject tabu moves unless aspiration applies.
6. Evaluate admissible moves using the common CVRPTW evaluator.
7. Accept the selected move, update Tabu memory, and update bandit rewards.
8. Return the best feasible or best penalized route plan found.

B. Comparison Methods

Several methods are implemented to evaluate the contribution of ABTS-RW. A constructive baseline represents fast route generation without advanced improvement. OR-Tools is included as an external CVRPTW solver under the common runtime budget. Fixed (F), roulette-adaptive (R), RL-guided (RL), and feasibility-safe sequential (FS) ALNS variants are used to compare against adaptive large-neighborhood search strategies. TS is used as the strongest complete classical baseline, while SA and ILS/VNS provide additional classical metaheuristic comparisons. A secondary adaptive variant, ABTS-RL, uses an RL-guided warm start followed by the same MAB-Tabu intensification phase.

C. Implementation Parameters and Reproducibility

The main implementation settings are summarized in Table II. Runtime budgets are assigned per method-instance-replication. The validation budget is 8 seconds for each 100-customer run. For the final scaled comparison, the budget is 11 seconds for each 150-customer run and 14 seconds for each 200-customer run. The same evaluator is used for all methods and measures driver-duty time, travel time, service time, waiting time, lateness, capacity violation, missing customers, duplicate visits, and feasibility.

TABLE II
EVALUATED METHODS AND IMPLEMENTATION SETTINGS

Component or method	Role	Settings used
Proposed ABTS-RW	Final proposed method	10% roulette warm start and 90% MAB-Tabu intensification. Operators: relocate-1, swap-1, Or-opt-2, intra-route 2-opt, and targeted relocate. Candidate moves per step: 80, 70, and 60; tabu tenures: 12, 14, and 18 for 100, 150, and 200 customer instances.
Common evaluator	Shared evaluation protocol	Driver-duty-time objective with penalties. Same evaluator for all methods.
Constructive	Initial heuristic and fast baseline	Capacity-aware and time-window-aware constructive route builder.
ABTS-RL	Secondary adaptive variant	Same MAB-Tabu intensification as ABTS-RW, but the first 10% of runtime uses an RL-guided warm start.
TS	Strongest complete classical baseline	Relocate and swap moves with probabilities 0.58 and 0.42. Candidate moves per iteration: 90 and 80; tabu tenures: 14 and 18 for 150 and 200 customer instances.
SA	Classical metaheuristic baseline	Neighbor moves are relocate, swap, 2-opt, and Or-opt with probabilities 0.34, 0.28, 0.24, and 0.14. Initial temperature: 0.015 times the initial objective value; final temperature: 0.002 times the initial temperature.
ILS/VNS	Classical metaheuristic baseline	Uses relocate, swap, 2-opt, and Or-opt neighborhoods. Initial local search samples 28 moves per neighborhood, stops after three non-improving rounds. Shaking strength increases from 2 to 8. Post-shaking local search samples 22 moves per neighborhood, stops after two non-improving rounds.

TABLE II (Cont.)

Component or method	Role	Settings used
ALNS variants	Adaptive large-neighborhood baselines	Destroy operators: random, worst, late, cluster, and route removal. Repair operators: greedy, regret-2, regret-3, time-oriented, and due-date-oriented insertion. Removal fractions are 0.08, 0.12, 0.18, and 0.25. F-ALNS uses fixed selection, while R-ALNS, RL-ALNS, and FS-ALNS adapt or sequence operator selection.
OR-Tools	External solver baseline	Uses parallel cheapest insertion as the first-solution strategy and guided local search as the local-search metaheuristic. Capacity and time dimensions are included, latest time windows use soft upper bounds, and route-span cost approximates driver-duty duration.

V. PERFORMANCE EVALUATION

A. Benchmark Dataset Construction

The benchmark dataset was prepared for Dammam using OpenStreetMap road-network and POI data extracted with OSMnx [17]. A directed drivable road graph was constructed, and candidate delivery locations were obtained from public map features representing retail and pharmacy-type points of interest. These locations were filtered and mapped to their nearest road-network nodes. Customer instances were then sampled from the candidate locations using multiple random seeds and problem sizes. For each sampled customer set, demands, service times, and time windows were assigned to represent delivery requirements. Deterministic travel-time matrices were generated for five operating scenarios: base, midday, morning peak, evening peak, and pessimistic. The spatial structure of the benchmark dataset is shown in Fig. 2.

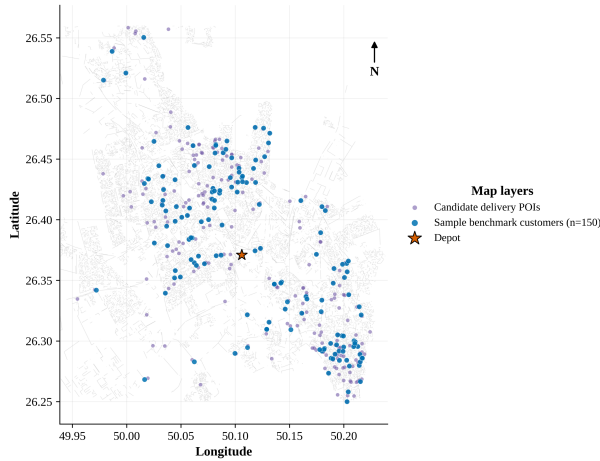


Fig. 2: Dammam road-network study area

B. Quantitative Analysis

The final evaluation focuses on held-out scaled-transfer instances with 150 and 200 customers under five congestion scenarios and two public seeds. For paired comparisons, improvement is computed on matched instance-replication pairs as $\frac{100(\text{reference} - \text{candidate})}{\text{reference}}$. Non-parametric Wilcoxon signed-rank tests, sign tests, pairwise wins/losses/ties, and bootstrap confidence intervals are used because heuristic routing performance

is stochastic and may not follow a normal distribution. OR-Tools is included as an external solver baseline. However, it returns solutions only for part of the scaled benchmark under the common runtime budget. Therefore, its objective values are interpreted together with solved-run and feasible-run coverage.

Table III shows the final 40 run comparison on the held-out scaled benchmark. Among the complete heuristic methods, ABTS-RW obtains the lowest mean objective. ABTS-RL is the closest adaptive competitor. TS is the strongest complete classical baseline. OR-Tools has a lower mean objective on its solved subset, but it solves only 20 of the 40 final runs.

TABLE III
FINAL 40 RUN COMPARISON ON THE HELD-OUT SCALED BENCHMARK

Method	Feasible runs	Mean objective (min)	Median objective (min)	Runtime (s)
Proposed ABTS-RW	39/40	6,804.58	6,584.46	13.44
ABTS-RL	39/40	6,839.50	6,644.91	13.22
Tabu Search	35/40	7,095.91	6,928.37	9.56
SA	37/40	7,228.17	6,995.87	12.50
ILS/VNS	33/40	7,524.94	7,282.14	12.50
FS-ALNS	36/40	7,328.18	6,968.63	14.46
RL-ALNS	33/40	7,300.60	7,048.89	13.18
R-ALNS	33/40	7,366.09	7,148.09	13.68
F-ALNS	34/40	7,670.14	7,303.72	12.68
OR-Tools	20/40	5,658.87	5,553.30	6.04
Constructive	31/40	8,061.99	7,771.32	0.05

The direct 40-run paired comparisons show that ABTS-RW improves over Tabu Search by 3.61% on average, with 36 wins and 4 losses. It improves over SA by 6.06%, with 36 wins and 4 losses, and over ILS/VNS by 9.80%, with 40 wins and no losses.

Table IV compares the top three methods over 100 paired runs. ABTS-RW achieves the lowest mean objective and the highest feasible-run coverage. ABTS-RL remains close, while TS has higher objective values and lower feasibility.

TABLE IV
TOP-THREE ROBUSTNESS COMPARISON OVER 100 PAIRED RUNS

Method	Feasible runs	Mean objective (min)	Median objective (min)	Mean lateness (min)	Runtime (s)
Proposed ABTS-RW	98/100	6,814.51	6,603.31	0.028	13.50
ABTS-RL	95/100	6,852.41	6,682.33	0.733	13.20
Tabu Search	87/100	7,116.52	6,965.17	1.396	9.50

ABTS-RW achieves a mean improvement of 3.81% and a median improvement of 3.31% against TS. It wins 94 of 100 paired comparisons, loses 6, and has no ties. Feasible-run coverage increases from 87/100 for TS to 98/100 for ABTS-RW. The Wilcoxon signed-rank test gives $p = 2.558 \times 10^{-17}$, and the sign test gives $p = 2.006 \times 10^{-21}$. The bootstrap 95% confidence interval for mean improvement is [3.25%, 4.40%], indicating that the improvement is statistically significant and not driven by a small number of favorable instances.

Performance across two groups (consisting of 150 & 200 customers) have been used to examine the scalability. ABTS-RW remains competitive at both sizes. For 150 customer group, ABTS-RW improves over TS by about 1.87% on

average and for 200 customer instances, the improvement increases to about 5.81%. This suggests that adaptive operator selection becomes more useful as the routing problem becomes larger and more constrained.

Table V shows the results of a validation-only sensitivity analysis conducted for the roulette warm-start fraction. ABTS family is stable across warm-start fractions as it is not dependent on a narrow parameter value. ABTS-RW is retained as the final method because it gives the best aggregate performance in the final held-out comparison and the 100-run robustness extension.

TABLE V
SENSITIVITY OF ABTS TO ROULETTE WARM-START FRACTION
ON VALIDATION INSTANCES

Configuration	Roulette warm-start fraction	Feasible runs	Mean objective (min)	Gap from best (%)	Runtime (s)
Proposed ABTS-RW	10%	20/20	3,920.91	0.355	8.26
No warm start	0%	20/20	3,907.03	0.000	8.04
Low warm start	5%	20/20	3,911.08	0.104	8.38
Medium warm start	15%	20/20	3,916.91	0.253	8.31
High warm start	20%	20/20	3,927.05	0.512	8.20

C. Qualitative Analysis

Fig. 3 provides a road-following route projection for the top methods on one held-out benchmark instance.

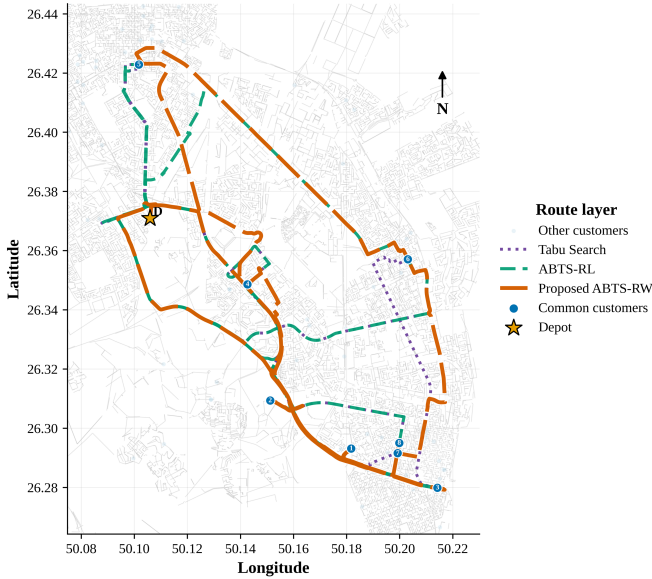


Fig. 3: Common customer road-following route projection

The main advantage of ABTS-RW is the combination of feasibility-aware neighborhood search with online adaptive operator selection. This allows the method to preserve the reliability of Tabu-style local search while learning which neighborhoods are useful during the current run. Even though ABTS-RL is a close competitor, it requires an additional RL-guided warm-start mechanism and still does not improve the

final aggregate result over ABTS-RW. TS remains a strong and fast complete classical baseline, but it has lower feasibility and higher objective values in the robustness extension. SA is simple and reasonably robust, but its objective quality is weaker than ABTS-RW. ILS/VNS provides useful multi-neighborhood local search, but it has lower feasibility and higher objective values. ALNS variants offer a flexible destroy-repair framework, but their performance is sensitive to operator and parameter design. OR-Tools is valuable as an external solver baseline and gives strong solutions on solved cases, but its limited solved-run coverage under the common runtime budget makes it less suitable as the main complete baseline.

VI. CONCLUSION

This study developed an adaptive heuristic framework for a last-mile delivery CVRPTW in Dammam, Saudi Arabia, using road-network travel times, customer demands, service times, time windows, vehicle capacity, and multiple congestion scenarios. The proposed ABTS-RW method addresses the problem by combining a short roulette-guided diversification phase with adaptive bandit-guided Tabu intensification, enabling the search to explore early and then focus on high-quality neighborhoods.

The results show that ABTS-RW provides the strongest overall performance among the complete heuristic methods considered in this study. In the 100-run robustness comparison, it improved over TS by 3.81% on average, won 94 out of 100 paired comparisons, and increased feasible-run coverage from 87/100 to 98/100. This improvement was supported by Wilcoxon and sign tests, as well as a bootstrap confidence interval. Additional comparisons against SA, ILS/VNS, ALNS variants, and the constructive baseline further confirmed the effectiveness of the proposed approach.

A practical challenge was that OR-Tools solved only part of the scaled benchmark within the common runtime budget, so its results had to be interpreted carefully. As stochastic metaheuristics also produce run-to-run variation, multiple replications and non-parametric tests were necessary to make the comparison reliable.

Overall, the proposed method is effective because it preserves feasibility-aware neighborhood search while adding online adaptive learning. Future work can refine the approach by extending the benchmark to larger instances, heterogeneous fleets, multiple depots, and dynamic order arrivals. It can also be strengthened by incorporating real-time traffic updates, parallel neighborhood evaluation, automated operator design, and integration with operational decision-support systems.

VII. DATA AND CODE AVAILABILITY

The benchmark datasets, implementation files, and experiment notebooks used in this study are publicly available at:

https://github.com/MynulVG/ISE_571_Heuristic_Search_Methods_g202528630_g202520390_Project

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